

AI-Powered Fake News Detection Using Text Classification

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Abstract— The rapid growth of social media and online news platforms has significantly increased the spread of fake news, making it difficult for users to identify reliable information. This project presents an AI-Powered Fake News Detection Using Text Classification system that automatically classifies news articles as real or fake using Machine Learning techniques. The dataset consists of 44,898 news articles, including 23,481 fake and 21,417 real news articles. The text data is preprocessed by converting it to lowercase, removing punctuation, numbers, URLs, and stop words to improve data quality. The cleaned text is then transformed into numerical features using the Term Frequency–Inverse Document Frequency (TF-IDF) technique. Four Machine Learning algorithms, namely Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and Neural Network (MLP Classifier), are trained and evaluated using performance metrics such as accuracy, precision, recall, and F1-score. Among all the models, Random Forest achieved the highest accuracy of 99.75%, making it the best-performing classifier. The proposed system provides an efficient solution for detecting fake news and can be extended for real-time and multilingual news verification.

Index Terms— Artificial Intelligence (AI), Fake News Detection, Machine Learning, Natural Language Processing (NLP), Text Classification, TF-IDF, Random Forest, Logistic Regression, K-Nearest Neighbors (KNN), Neural Network (MLP).

I. INTRODUCTION

A. Background

The rapid growth of the internet and social media has made news easily accessible to millions of users. However, this has also increased the spread of fake news, which contains false or misleading information. Fake news can influence public opinion, create

confusion, and spread misinformation on topics such as politics, health, and finance. Since manually verifying every news article is difficult and time-consuming, automated fake news detection systems have become increasingly important.

B. Problem Statement

Detecting fake news manually is challenging because of the large volume of news published every day. Traditional verification methods are slow and require human effort. Therefore, there is a need for an intelligent system that can automatically classify news articles as real or fake using Machine Learning techniques.

C. Objectives

- Develop an AI-based fake news detection system.
- Preprocess and clean textual news data.
- Convert text into numerical features using TF-IDF.
- Train and compare multiple Machine Learning algorithms.
- Identify the best-performing model for fake news detection.

II. LITERATURE REVIEW

Researchers have proposed various methods to detect fake news using Natural Language Processing (NLP) and Machine Learning. Traditional algorithms such as Logistic Regression, Naive Bayes, and Support Vector Machines have shown good performance for text classification. Ensemble learning methods such as Random Forest improve prediction accuracy by combining multiple decision trees. Recent approaches also use Deep Learning models such as

LSTM and BERT for understanding complex language patterns. Although these methods achieve high accuracy, they often require large datasets and computational resources. This project compares multiple Machine Learning models and selects the most suitable one based on performance.

III. PROBLEM STATEMENT

The increasing amount of fake news shared on digital platforms makes it difficult for users to identify trustworthy information. Manual verification is time-consuming and cannot handle the volume of online content. The proposed system uses Artificial Intelligence and Machine Learning to automatically classify news articles as real or fake based on their textual content.

IV. SYSTEM REQUIREMENT

A. Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4GB or higher
- Storage: 10GB free space

B. Software Requirements

- Windows 10/11
- Python 3.x
- Jupyter Notebook
- Anaconda
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- NLTK

V. DATASET DESCRIPTION

The project uses the Kaggle Fake and Real News Dataset.

Dataset Details:

- FakeNewsRecords: **23,481**
- RealNewsRecords: **21,417**

- TotalRecords: **44,898**

Dataset Features:

Column	Description
Title	News headline
Text	News content
Subject	News category
Date	Publication date

Labels:

- FakeNews=0
- RealNews=1

VI. PROPOSED METHODOLOGY

The project follows these steps:

1. Import libraries.
2. Load the Fake and Real datasets.
3. Clean the text by removing punctuation, numbers, URLs, and stopwords.
4. Merge the datasets.
5. Shuffle the combined dataset.
6. Convert text into TF-IDF vectors.
7. Split the dataset into training and testing sets.
8. Train Logistic Regression, KNN, Random Forest, and Neural Network models.
9. Evaluate each model using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.
10. Compare model performance and select the best model.

VII. ALGORITHMS USED

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Random Forest
- Neural Network (MLP Classifier)

VIII. EXPERIMENTAL RESULTS

Model	Accuracy
LogisticRegression	98.68%
KNN	70.22%
RandomForest	99.75%
NeuralNetwork	99.13%

Random Forest achieved the highest accuracy and provided the best classification performance.

IX. CONCLUSION

This project successfully developed an AI-powered fake news detection system using Machine Learning techniques. After preprocessing the dataset and converting text into TF-IDF features, four classification algorithms were trained and evaluated. Random Forest achieved the highest accuracy of **99.75%**, making it the most effective model for detecting fake news. The system can assist users in identifying misleading information and contributes to reducing the spread of misinformation.

X. FUTURE SCOPE

- Real-time news verification.
- Integration with social media platforms.
- Multilingual fake news detection.
- Deep Learning models such as BERT.
- Mobile and web application deployment.

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